

Edge Fault Tolerance for Two-Pair Spanning Disjoint Paths of Star Networks.

Chun-Nan Hung, Chi-Lai Liu and Hsuan-Han Chang
 Department of Computer Science and Information Engineering
 Dayeh University
 {spring,r9506036,r9306019}@mail.dyu.edu.tw

Abstract

The star graph is a famous interconnection networks. In this paper, we discuss the edge fault tolerance for 2-pair spanning property for star graph. Let $S_n = (V_1 \cup V_2, E)$ be the n -dimensional star graph. Let U be the set of any k pairs of vertices of $G = (V_b \cup V_w, E)$ for $|U \cap V_b| = |U \cap V_w| = k$. The graph $G = (V_b \cup V_w, E)$ is k -pair spanning if there exist k spanning disjoint paths between every pair of vertices of U . In this paper, we will investigate the edge fault tolerance for 2-pair spanning property of star graph. We will show that $S_n - F_e$ is 2-pair spanning where F_e is the set of faulty edge for $|F_e| \leq n - 4$ and $n \geq 5$. The number of faulty edges is optimal.

Keywords: star graph; fault tolerance; spanning disjoint paths.

1 Introduction

The multiprocessor systems and computer networks are connected based on interconnection networks. Popular instances of interconnection networks include hypercube, star graph, etc. The star graph (introduced in [1]) has been recognized as an attractive alternative to the popular hypercubes. The hypercube and star graph have many advantage topological properties such as symmetry, fault tolerance, maximal fault tolerance and diameter.

The interconnection network is common expressed as an undirected graph. The vertices on the graph represent processors and the edges represent links between processors. The Hamiltonian path is a path that visits every vertex exactly once. A bipartite graph $G = (V_1 \cup V_2, E)$ is Hamiltonian laceable if there exists a Hamil-

tonian path between every pair of vertices x and y with $x \in V_1$ and $y \in V_2$. A bipartite graph $G = (V_1 \cup V_2, E)$ is k edges Hamiltonian laceable if $G - F_e$ is Hamiltonian laceable for $F_e \subset E$ and $|F_e| \leq k$. The star graph $S_n = (V_1 \cup V_2, E)$ and hypercube $Q_n = (V_b \cup V_w, E)$ are both bipartite graphs. In [3], Li et al. proved that S_n is $(n - 3)$ edges Hamiltonian laceable. The hypercube Q_n is $(n - 2)$ edges Hamiltonian laceable shown in [6].

Let P be a path and $V(P)$ be the set of vertices on P . Two paths P_1 and P_2 are *two spanning disjoint paths* of $G = (V, E)$ if $V(P_1) \cap V(P_2) = \emptyset$ and $V(P_1) \cup V(P_2) = V$. In [5], the authors showed that for any $s_1, s_2 \in V_b$ and $t_1, t_2 \in V_w$ there exist two spanning disjoint paths $P(s_1, t_1)$ and $P(s_2, t_2)$ of Q_n . However, there is no paper about two spanning disjoint paths for star graphs.

In [2], Hung et al. proposed the multiple spanning disjoint paths for hypercube. Let U be the set of any k pairs of vertices of $G = (V_b \cup V_w, E)$ for $|U \cap V_b| = |U \cap V_w| = k$. The graph $G = (V_b \cup V_w, E)$ is k -pair spanning if there exist k spanning disjoint paths between every pair of vertices of U . Hung et al. showed that Q_n is $\lfloor \frac{n}{2} \rfloor$ -pair spanning for $n \geq 2$ in [2]. We can't find any papers about k -pair spanning disjoint paths for star graph for $k \geq 2$.

In this paper, we will investigate the edge fault tolerance for 2-pair spanning property of star graph. We will show that $S_n - F_e$ is 2-pair spanning where F_e is the set of faulty edge for $|F_e| \leq n - 4$ and $n \geq 5$.

The rest of this paper is organized as follows. In next section, we brief introduce properties of star graph. The main results are shown in section 3. Finally, we present the conclusion in section 4.

2 Definition and Basic Properties

Definition 1 The n -dimensional star graph is denoted by S_n . The vertex set $V = \{v | v \text{ is a permutation of } 1, 2, \dots, n\}$ and edge set $E = \{(v_1 v_2 \dots v_{i-1} v_i \dots v_n, v_i v_2 \dots v_{i-1} v_1 v_{i+1} \dots v_n) | v_1 \dots v_n \in V \text{ and } 2 \leq i \leq n\}$

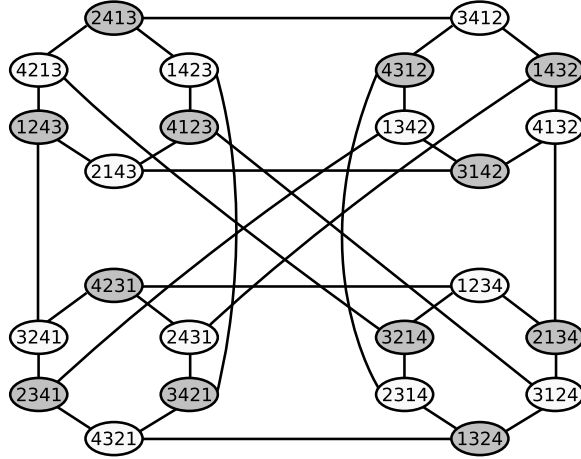


Figure 1: Illustration of S_4 .

For example, vertex 1234 in S_4 connects to 2134, 4231, 3214 as illustrated in Figure 1.

A star graph S_n is an edge symmetric and node symmetric graph. By definition, S_n contains $n!$ vertices and $\frac{n!(n-1)}{2}$ edges. Each vertex of star graph is $(n-1)$ regular. The degree of a vertex is the number of edges adjacent to v denote $d_G(v)$. The regular degree denote every node is incident with the same number of edges.

Let $S_n = (V_1 \cup V_2, E)$ be the n -dimensional star network where V_1 and V_2 are the set of black vertices and white vertices, respectively. The graph S_n can be decomposed to n copies of $(n-1)$ -dimensional star graphs. Give $2 \leq k \leq n$, we can use S_{n-1}^i to denote the subgraph of S_n induced by the vertices whose k th digit is i . Thus, S_{n-1}^i is isomorphic to the $(n-1)$ -dimensional star graph S_{n-1} for $1 \leq i \leq n$. Let N_n be the set of $\{1, 2, \dots, n\}$, and M_n be any nonempty subset of N_n . Let $S_{n-1}^{M_n}$ be the subgraphs of S_n induced by $\bigcup_{i \in M_n} V(S_{n-1}^i)$. Let F_e denote the set of faulty edges of S_n and $f_e = |F_e|$. We also use F_e^i to denote the set of faulty edges in S_{n-1}^i and $f_e^i = |F_e^i|$. The F_e^C is set of faulty edge between S_{n-1}^k and S_{n-1}^j for $1 \leq k < j \leq n$. Let v be a vertex of S_{n-1}^i for some $1 \leq i \leq n$. Let $E_e(v)$ be the edge set $\{(u, v) | \text{for } u \in V(S_{n-1}^i) \text{ and } (u, u') \in F_e\}$. Thus,

$|E_e(v)| + |F_e^i| \leq |F_e| \leq n-4$. For every vertex $v \in V(S_{n-1}^i)$, v' is the only neighbor of v in S_{n-1}^j for $1 \leq i \neq j \leq n$.

Li et al. proved the follow lemma in [3].

Lemma 1 The graph S_n is $(n-3)$ -edge Hamiltonian laceable with $n \geq 4$.

In [4], Lin et al. proved the following lemma.

Lemma 2 If x and $y \in S_{n-1}^i$ which $1 \leq i \leq n$, and $d(x, y) \leq 2$, then $n(x') \neq n(y')$

The following lemma is the same as **Lemma 8** in [3].

Lemma 3 Let M_n be any nonempty subset of N_n for $|M_n| \geq 2$ and i_1, i_2 be two different elements of M_n . For $n \geq 5$ and $|F_e^i| \leq (n-4)$, any $s \in (V_1 \cap V(S_{n-1}^{i_1}))$ and $t \in (V_2 \cap V(S_{n-1}^{i_2}))$, there exists a Hamiltonian path between s and t in $S_{n-1}^{M_n} - (\bigcup_{i \in M_n} F_e^i)$.

3 Main result

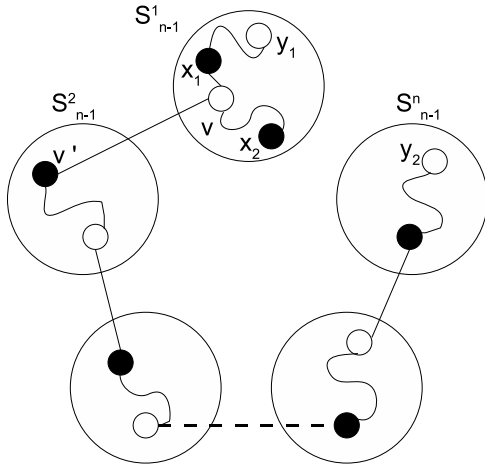
Lemma 4 Let $S_n = (V_1 \cup V_2, E)$ be an n -dimensional star graph and F_e be the set of faulty edges of S_n . For any $x_1, x_2 \in V_1, y_1, y_2 \in V_2$, there exist two spanning disjoint paths $P(x_1, y_1)$ and $P(x_2, y_2)$ in $S_n - F_e$ with $|F_e| \leq (n-4)$ and $n \geq 5$.

Proof:

There exists some k , for $2 \leq k \leq n$, such that the k th digits of y_1 and y_2 are different. We can use digit k to divide S_n into n subgraphs S_{n-1}^i for $1 \leq i \leq n$. Without loss of generality, we can assume that $k = n$. Thus, y_1 and y_2 are in different subgraphs. Without loss of generality, we can assume that $y_1 \in V(S_{n-1}^1)$ and $y_2 \in V(S_{n-1}^n)$. We will construct the spanning disjoint paths $P(x_1, y_1)$ and $P(x_2, y_2)$ in $S_n - F_e$ in the following cases.

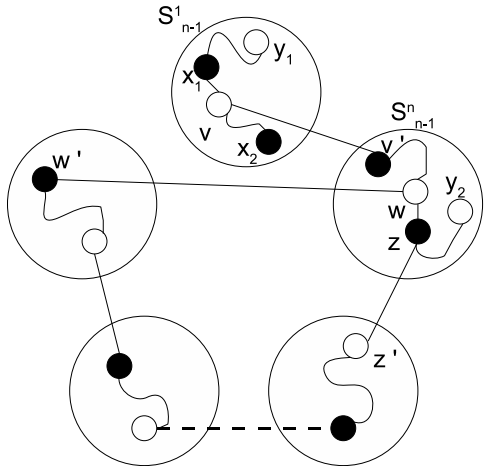
Case 1: $x_1, x_2 \in V(S_{n-1}^1)$.

Applying Lemma 1, we can obtain that there exists a Hamiltonian path P between y_1 and x_2 of $S_{n-1}^1 - F_e - E_e(x_1)$. Thus, $(v, v') \notin F_e$. We can denote P as $\langle x_2 \rightarrow P(x_2, v) \rightarrow v, x_1 \rightarrow P(x_1, y_1) \rightarrow y_1 \rangle$. We will discuss the following sub-cases according to the location of v' .


 Figure 2: Illustration of **Case 1.1**.

Case 1.1: $v' \notin V(S_{n-1}^n)$.

Without loss of generality, we can assume that $v' \in V(S_{n-1}^2)$. Applying Lemma 3, we can obtain that there exists a Hamiltonian path $P(v', y_2)$ in $S_{n-1}^{\{2,3,\dots,n\}}$. Thus, $P(x_1, y_1)$ and $\langle x_2 \rightarrow P(x_2, v) \rightarrow v, v' \rightarrow P(v', y_2) \rightarrow y_2 \rangle$ form two spanning disjoint paths in $S_n - F_e$, as illustrated in Figure 2.


 Figure 3: Illustration of **Case 1.2**.

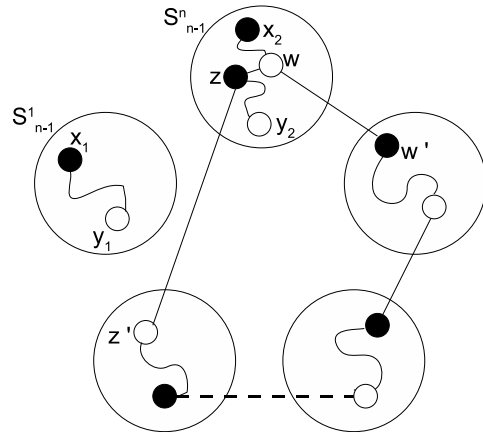
Case 1.2: $v' \in V(S_{n-1}^n)$.

Applying Lemma 1, there exists a Hamiltonian path $P(v', y_2)$ of $S_{n-1}^n - F_e$. Since $n \geq 5$, there exists an edge (w, z) on $P(v', y_2)$ such that $w', z' \notin V(S_{n-1}^1)$ and $(w, w'), (z, z') \notin F_e$. We denote $P(v', y_2)$ as $\langle v' \rightarrow P(v', w) \rightarrow w, z \rightarrow$

$P(z, y_2) \rightarrow y_2 \rangle$. Applying Lemma 2, we know $w' \in V(S_{n-1}^{i_1})$ and $z' \in V(S_{n-1}^{i_2})$ for $2 \leq i_1 \neq i_2 \leq n-1$. Applying Lemma 3, we can obtain that there exists a Hamiltonian path $P(w', z')$ in $S_{n-1}^{\{2,3,\dots,n-1\}}$. Thus, $P(x_1, y_1)$ and $\langle x_2 \rightarrow P(x_2, v) \rightarrow v, v' \rightarrow P(v', w) \rightarrow w, w' \rightarrow P(w', z') \rightarrow z', z \rightarrow P(z, y_2) \rightarrow y_2 \rangle$ form two spanning disjoint paths in $S_n - F_e$, as illustrated in Figure 3.

Case 2: $x_1 \in V(S_{n-1}^1)$ and $x_2 \in V(S_{n-1}^n)$.

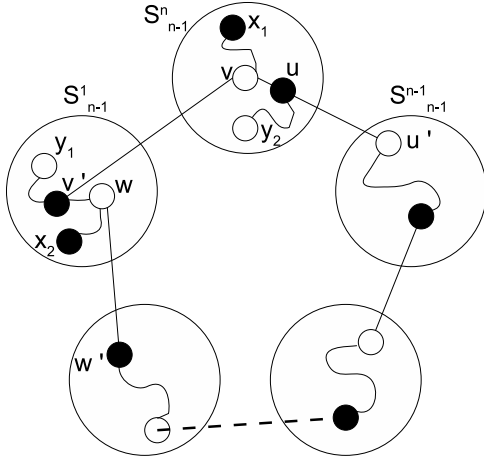
Applying Lemma 1, we can find Hamiltonian paths $P(x_1, y_1)$ and $P(x_2, y_2)$ in $S_{n-1}^1 - F_e^1$ and $S_{n-1}^n - F_{n-1}^n$, respectively. Since $n \geq 5$, there exists an edge (w, z) on $P(x_2, y_2)$ such that $w', z' \notin V(S_{n-1}^1)$ and $(w, w'), (z, z') \notin F_e$. We denote $P(x_2, y_2)$ as $\langle x_2 \rightarrow P(x_2, w) \rightarrow w, z \rightarrow P(z, y_2) \rightarrow y_2 \rangle$. Applying Lemma 3, we can obtain that there exists a Hamiltonian path $P(w', z')$ in $S_{n-1}^{\{2,3,\dots,n-1\}}$. Thus, $P(x_1, y_1)$ and $\langle x_2 \rightarrow P(x_2, w) \rightarrow w, w' \rightarrow P(w', z') \rightarrow z', z \rightarrow P(z, y_2) \rightarrow y_2 \rangle$ form two spanning disjoint paths in $S_n - F_e$, as illustrated in Figure 4.


 Figure 4: Illustration of **Case 2**.

Case 3: $x_1 \in V(S_{n-1}^n)$ and $x_2 \in V(S_{n-1}^1)$.

Without loss of generality, we can assume that $f_e^1 \leq f_e^n$. Applying Lemma 1, we can find a Hamiltonian path $P(x_1, y_2)$ in $S_{n-1}^n - F_e^n$. Let (v, u) be an edge on $P(x_1, y_2)$ for $(u, u'), (v, v') \notin F_e$, $v' \in V(S_{n-1}^1)$ and $f_e^1 + |E_e(v')| \leq n-5$. We denote $P(x_1, y_2)$ as $\langle x_1 \rightarrow P(x_1, v) \rightarrow v, u \rightarrow P(u, y_2) \rightarrow y_2 \rangle$. We can assume that $u' \in V(S_{n-1}^{n-1})$. Let (v', z) be the edge for $z' \in V(S_{n-1}^{n-1})$. Since

$f_e^1 \leq f_e^n$, $f_e^1 + |E_e(v')| \leq n - 5$. Thus, $S_{n-1}^1 - F_e^1 - E_e(v') - \{(v', z)\}$ is Hamiltonian laceable. There exists a Hamiltonian path $\langle x_2 \rightarrow P(x_2, w) \rightarrow w, v' \rightarrow P(v', y_1) \rightarrow y_1 \rangle$ in $S_{n-1}^1 - F_e^1 - E_e(v') - \{(v', z)\}$. Applying Lemma 3, we can obtain that there exists a Hamiltonian path $P(w', u')$ in $S_{n-1}^{\{2,3,\dots,n-1\}}$. Therefore, $\langle x_1 \rightarrow P(x_1, v) \rightarrow v, v' \rightarrow P(v', y_1) \rightarrow y_1 \rangle$ and $\langle x_2 \rightarrow P(x_2, w) \rightarrow w, w' \rightarrow P(w', u') \rightarrow u', u \rightarrow P(u, y_2) \rightarrow y_2 \rangle$ form two spanning disjoint paths in $S_n - F_e$, as illustrated in Figure 5.

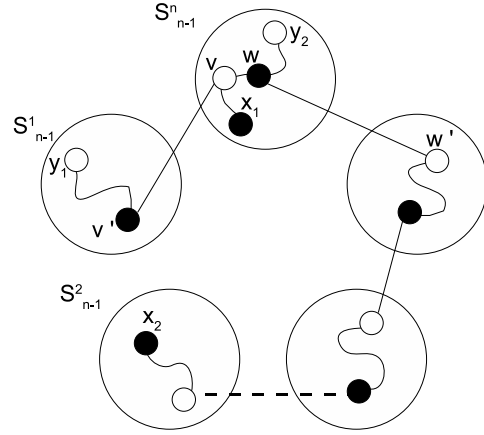

 Figure 5: Illustration of **Case 3**.

Case 4: $x_1 \in V(S_{n-1}^1)$ and $x_2 \notin V(S_{n-1}^1 \cup S_{n-1}^n)$. Applying Lemma 1, we can obtain that there exists a Hamiltonian path P between x_1 and y_1 of $S_{n-1}^1 - F_e^1$. Applying Lemma 3, we can obtain that there exists a Hamiltonian path $P(x_2, y_2)$ in $S_{n-1}^{\{2,3,\dots,n\}} - F_e$. These are two spanning disjoint paths of $S_n - F_e$.

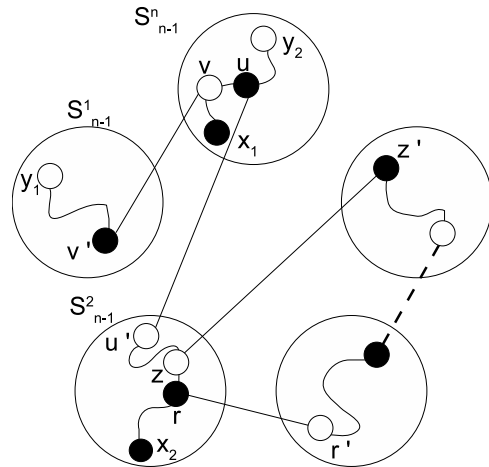
Case 5: $x_1 \in V(S_{n-1}^n)$ and $x_2 \notin V(S_{n-1}^1 \cup S_{n-1}^n)$. Without loss of generality, we can assume that $x_2 \in V(S_{n-1}^2)$.

Case 5.1: $|F_e^n| \leq n - 5$

Let (v, u) be an edge of S_{n-1}^n such that v is a white vertex and $v' \in V(S_{n-1}^1)$, $u' \in V(S_{n-1}^2)$, $(v, v') \notin F_e$ and $|E_e(v)| + |F_e^n| \leq n - 5$. Applying Lemma 1, we can obtain that there exists a Hamiltonian path $P(x_1, y_2)$ of $S_{n-1}^n - F_e^n - E_e(v) - \{(v, u)\}$. We denote $P(x_1, y_2)$ as $\langle x_1 \rightarrow P(x_1, v) \rightarrow v, w \rightarrow P(w, y_2) \rightarrow y_2 \rangle$. Applying Lemma 1, we can obtain that there exists a Hamiltonian path $P(v', y_1)$


 Figure 6: Illustration of **Case 5.1**.

of $S_{n-1}^1 - F_e^1$. Applying Lemma 3, we can construct the Hamiltonian path $P(x_2, w')$ in $S_{n-1}^{\{2,3,\dots,n-1\}} - F_e$. Thus, $\langle x_1 \rightarrow P(x_1, v) \rightarrow v, v' \rightarrow P(v', y_1) \rightarrow y_1 \rangle$ and $\langle x_2 \rightarrow P(x_2, w') \rightarrow w', w \rightarrow P(w, y_2) \rightarrow y_2 \rangle$ form two spanning disjoint paths of $S_n - F_e$, as illustrated in Figure 6.


 Figure 7: Illustration of **Case 5.2**.

Case 5.2: $|F_e^n| = n - 4$

Let (v, u) be an edge of S_{n-1}^n such that v is a white vertex and $v' \in V(S_{n-1}^1)$, $u' \in V(S_{n-1}^2)$ and $(v, v') \notin F_e$. Applying Lemma 1, we can construct the Hamiltonian path $P(x_1, y_2)$ of $S_{n-1}^n - F_e^n$. We denote $P(x_1, y_2)$ as $\langle x_1 \rightarrow P(x_1, v) \rightarrow v, u \rightarrow P(u, y_2) \rightarrow y_2 \rangle$. When $w \neq u$, the two spanning disjoint

paths can be constructed as the same as **Case 5.1**. In the follows, we will construct the two spanning disjoint paths for $w = u$. Let z be a vertex of S_{n-1}^2 and $z' \notin V(S_{n-1}^1 \cup S_{n-1}^n)$. Let (z, z_1) and (z, z_2) be two edges of S_{n-1}^2 for $z_1 \in V(S_{n-1}^1), z_2 \in V(S_{n-1}^n)$. Applying Lemma 1, we can construct the Hamiltonian path $P(x_2, u') = \langle x_2 \rightarrow P(x_2, r) \rightarrow r, z \rightarrow P(z, u') \rightarrow u' \rangle$ of S_{n-1}^2 . Applying Lemma 3, we can construct the Hamiltonian path $P(r', z')$ in $S_{n-1}^{\{3, \dots, n-1\}}$. Therefore, $\langle x_1 \rightarrow P(x_1, v) \rightarrow v, v' \rightarrow P(v', y_1) \rightarrow y_1 \rangle$ and $\langle x_2 \rightarrow P(x_2, r) \rightarrow r, r' \rightarrow P(r', z') \rightarrow z', z \rightarrow P(z, u') \rightarrow u', u \rightarrow P(u, y_2) \rightarrow y_2 \rangle$ form two spanning disjoint paths of $S_n - F_e$, as illustrated in Figure 7.

Case 6: $x_1, x_2 \notin V(S_{n-1}^1 \cup S_{n-1}^n)$.

Case 6.1: $x_1 \in V(S_{n-1}^{i_1}), x_2 \in V(S_{n-1}^{i_2})$ for some $2 \leq i_1 \neq i_2 \leq n-1$

Without loss of generality, we can assume that $x_1 \in V(S_{n-1}^2)$ and $x_2 \in V(S_{n-1}^3)$. Let v be a vertex of S_{n-1}^2 for $v' \in V(S_{n-1}^1)$ and $(v, v') \notin F_e$. Applying Lemma 1, we can obtain that there exist Hamiltonian paths $P(x_1, v)$ and $P(v', y_1)$ of $S_{n-1}^2 - F_e^2$ and $S_{n-1}^1 - F_e^1$, respectively. Applying lemma 3, we can find a Hamiltonian path $P(x_2, y_2)$ in $V(S_{n-1}^{3,4, \dots, n})$. Thus, $\langle x_1 \rightarrow P(x_1, v) \rightarrow v, v' \rightarrow P(v', y_1) \rightarrow y_1 \rangle$ and $P(x_2, y_2)$ form two spanning disjoint paths of $S_n - F_e$, as illustrated in Figure 8.

Case 6.2: $x_1, x_2 \in V(S_{n-1}^i)$ for some $2 \leq i \leq n-1$

Without loss of generality, we can assume that $x_1, x_2 \in V(S_{n-1}^2)$ and $f_e^1 \geq f_e^n$.

Case 6.2-1: $f_e^2 + |E_e(x_1)| \leq n-5$

Let $(v, v') \notin F_e$ be an edge for $v \in V(S_{n-1}^2), v' \in V(S_{n-1}^1)$. Applying Lemma 1, we can obtain Hamiltonian paths $P(y_1, v')$ in $S_{n-1}^1 - F_e^1$ and $\langle x_2 \rightarrow P(x_2, z) \rightarrow z, x_1 \rightarrow P(x_1, v) \rightarrow v \rangle$ in $S_{n-1}^2 - F_e^2$ for $z' \notin V(S_{n-1}^1)$.

Suppose that $z' \notin V(S_{n-1}^n)$. Applying Lemma 3, we can obtain a Hamiltonian path $P(z', y_2)$ in $S_{n-1}^{\{3, \dots, n\}} - F_e$. Therefore, $\langle x_1 \rightarrow P(x_1, v) \rightarrow v, v' \rightarrow P(v', y_1) \rightarrow y_1 \rangle$ and $\langle x_2 \rightarrow P(x_2, z) \rightarrow z, z' \rightarrow P(z', y_2) \rightarrow y_2 \rangle$ form two spanning disjoint paths of $S_n - F_e$, as illustrated in Figure 10.

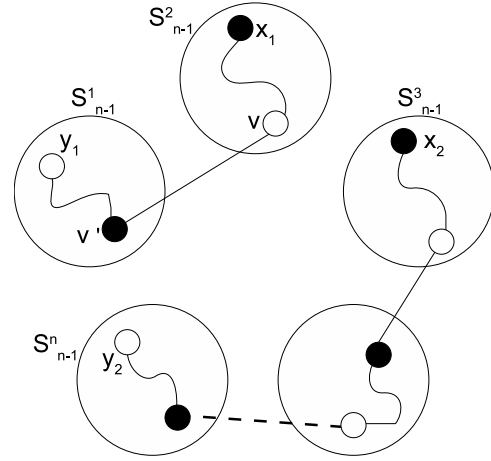


Figure 8: Illustration of **Case 6.1**.

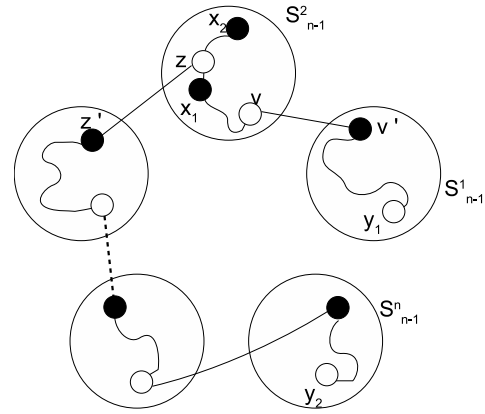


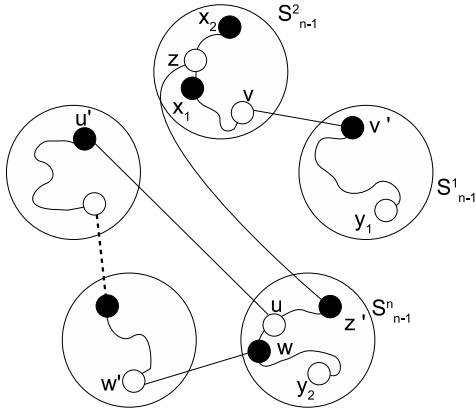
Figure 9: Illustration of **Case 6.2-1**.

$z, z' \rightarrow P(z', y_2) \rightarrow y_2$ form two spanning disjoint paths of $S_n - F_e$, as illustrated in Figure 10.

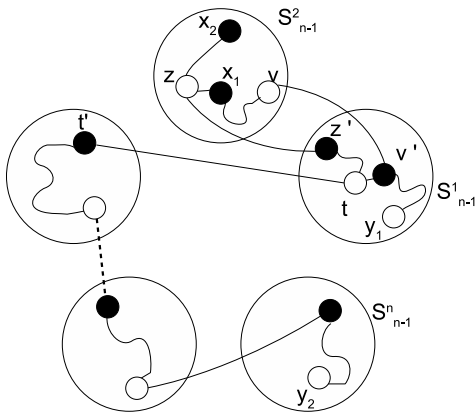
Suppose that $z' \in V(S_{n-1}^n)$. Let (u, w) be an edge of S_{n-1}^n for $(u, u'), (w, w') \notin F_e$ and $u', w' \notin V(S_{n-1}^1 \cup S_{n-1}^2)$. Since $f_e^n \leq f_e^1$, there exists a Hamiltonian path $\langle z' \rightarrow P(z', u) \rightarrow u, w \rightarrow P(w, y_2) \rightarrow y_2 \rangle$ of $S_{n-1}^n - F_e^n$. Applying Lemma 3, we can obtain a Hamiltonian path $P(u', w')$ in $S_{n-1}^{\{3, \dots, n-1\}} - F_e$. Therefore, $\langle x_1 \rightarrow P(x_1, v) \rightarrow v, v' \rightarrow P(v', y_1) \rightarrow y_1 \rangle$ and $\langle x_2 \rightarrow P(x_2, z) \rightarrow z, z' \rightarrow P(z', u) \rightarrow u, u' \rightarrow P(u', w') \rightarrow w', w \rightarrow P(w, y_2) \rightarrow y_2 \rangle$ form two spanning disjoint paths of $S_n - F_e$, as illustrated in Figure 10.

Case 6.2-2: $f_e^2 + |E_e(x_1)| = n-4$

Let $(v, v') \notin F_e$ be an edge for $v \in$


 Figure 10: Illustration of **Case 6.2-1**.

$V(S_{n-1}^2), v' \in V(S_{n-1}^1)$. Applying Lemma 1, we can obtain a Hamiltonian path $\langle x_2 \rightarrow P(x_2, z) \rightarrow z, x_1 \rightarrow P(x_1, v) \rightarrow v \rangle$ in $S_{n-1}^2 - F_e^2$. When $z' \notin V(S_{n-1}^1)$, we can construct the two spanning disjoint paths as same as **Case 6.2-1**. We will discuss the condition $z' \in V(S_{n-1}^1)$ in the follows. Since $f_e^2 + |E_e(x_1)| = n - 4$, $f_e^1 = 0$. Applying Lemma 1, we can construct a Hamiltonian path $\langle y_1 \rightarrow P(y_1, v') \rightarrow v', t \rightarrow P(t, z') \rightarrow z' \rangle$ in S_{n-1}^1 for $t' \notin V(S_{n-1}^n)$. Applying Lemma 3, we can obtain a Hamiltonian path $P(t', y_2)$ in $S_{n-1}^{\{3, \dots, n\}}$. Therefore, $\langle x_1 \rightarrow P(x_1, v) \rightarrow v, v' \rightarrow P(v', y_1) \rightarrow y_1 \rangle$ and $\langle x_2 \rightarrow P(x_2, z) \rightarrow z, z' \rightarrow P(z', t) \rightarrow t, t' \rightarrow P(t', y_2) \rightarrow y_2 \rangle$ form two spanning disjoint paths of $S_n - F_e$, as illustrated in Figure 11. $\square$


 Figure 11: Illustration of **Case 6.2-3**.

Lemma 5 Let $S_n = (V_1 \cup V_2, E)$ be an n -dimensional star graph and F_e be the set of faulty edges of S_n . Let $x_1, y_1 \in V_1$ and $x_2, y_2 \in V_2$ be arbitrary vertices of S_n . There exist two spanning disjoint paths $P(x_1, y_1)$ and $P(x_2, y_2)$ in $S_n - F_e$ for $|F_e| \leq (n - 4)$ and $n \geq 5$.

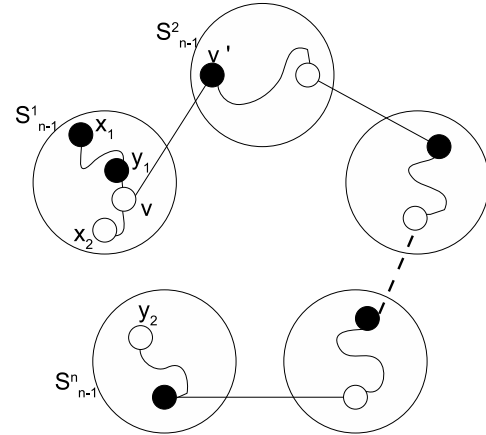
Proof:

Since the star graph is symmetric, we only need to consider the condition x_2 and y_2 are in different subgraphs. Without loss of generality, we can assume that $x_2 \in V(S_{n-1}^1)$ and $y_2 \in V(S_{n-1}^n)$. We will construct the spanning disjoint paths $P(x_1, y_1)$ and $P(x_2, y_2)$ of $S_n - F_e$ with the following cases.

Case 1: At least one of x_1, y_1 is in $S_{n-1}^1 \cup S_{n-1}^n$. Without loss of generality, we can assume that $x_1 \in V(S_{n-1}^1)$.

Case 1.1: $y_1 \in V(S_{n-1}^1)$

Applying Lemma 1, we can obtain that there exists a Hamiltonian path P between x_1 and x_2 of $S_{n-1}^1 - F_e - E_e(y_1)$. Thus, $(v, v') \notin F_e$. We can denote P as $\langle x_2 \rightarrow P(x_2, v) \rightarrow v, y_1 \rightarrow P(y_1, x_1) \rightarrow x_1 \rangle$. We will discuss the following subcases according to the location of v' .


 Figure 12: Illustration of **Case 1.1-1**.

Case 1.1-1: $v' \notin V(S_{n-1}^n)$.

Without loss of generality, we can assume that $v' \in V(S_{n-1}^2)$. Applying Lemma 3, we can obtain that there exists a Hamiltonian path $P(v', y_2)$ in $S_{n-1}^{\{2, 3, \dots, n\}}$. Thus, $P(x_1, y_1)$ and $\langle x_2 \rightarrow P(x_2, v) \rightarrow v, v' \rightarrow P(v', y_2) \rightarrow y_2 \rangle$ form two spanning disjoint paths in $S_n - F_e$, as illustrated in Figure 12.

1.3-1. In the follows, we will construct two spanning disjoint paths for $v = z_1$. Let (w, u) be an edge of S_{n-1}^{n-1} for $(w, w'), (u, u') \notin F_e$ and $w', u' \notin V(S_{n-1}^1 \cup S_{n-1}^n)$. Applying Lemma 1, we can construct the Hamiltonian path $P(v', y_1) = \langle v' \rightarrow P(v', w) \rightarrow w, u \rightarrow P(u, y_1) \rightarrow y_1 \rangle$ of S_{n-1}^{n-1} . Applying Lemma 3, we can construct the Hamiltonian path $P(w', u')$ in $S_{n-1}^{\{2,3,\dots,n-2\}}$. Therefore, $\langle x_2 \rightarrow P(x_2, z) \rightarrow z, z' \rightarrow P(z', y_2) \rightarrow y_2 \rangle$ and $\langle x_1 \rightarrow P(x_1, v) \rightarrow v, v' \rightarrow P(v', w) \rightarrow w, w' \rightarrow P(w', u') \rightarrow u', u \rightarrow P(u, y_1) \rightarrow y_1 \rangle$ form two spanning disjoint paths of $S_n - F_e$, as illustrated in Figure 16.

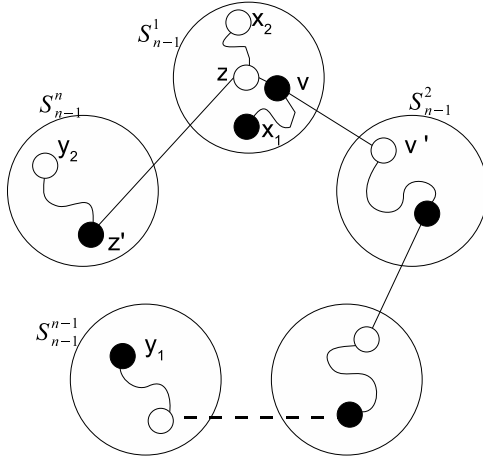


Figure 15: Illustration of **Case1.3-1**.

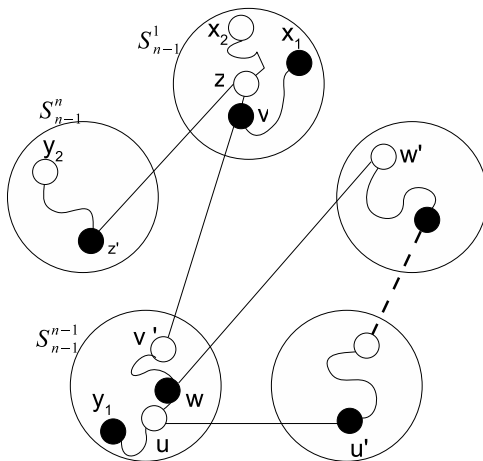


Figure 16: Illustration of **Case1.3-2**.

Case 2: $x_1, y_1 \notin V(S_{n-1}^1 \cup S_{n-1}^n)$.

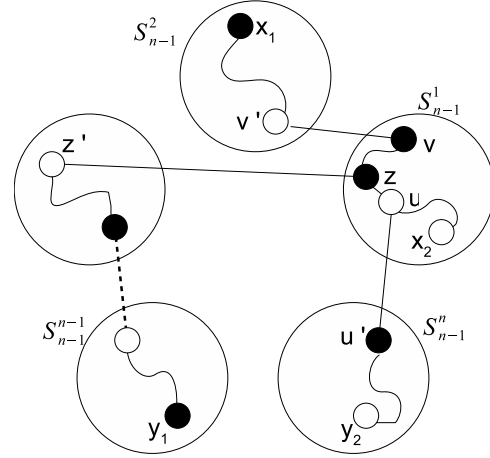


Figure 17: Illustration of **Case2.1**.

Case 2.1: $x_1 \in V(S_{n-1}^{k_1}), y_1 \in V(S_{n-1}^{k_2})$ for some $2 \leq k_1 \neq k_2 \leq n-1$

Without loss of generality, we can assume that $x_1 \in S_{n-1}^2, y_1 \in V(S_{n-1}^{n-1})$ and $f_e^1 \leq f_e^n$. Let u and v be two vertices of S_{n-1}^1 for $(u, u'), (v, v') \notin F_e$ and $u' \in V(S_{n-1}^n), v' \in V(S_{n-1}^2)$. Let (u, u_1) be an edge of S_{n-1}^1 for $u_1 \in V(S_{n-1}^2)$. Applying Lemma 1, we can construct a Hamiltonian path $P(x_2, v) = \langle x_2 \rightarrow P(x_2, u) \rightarrow u, z \rightarrow P(z, v) \rightarrow v \rangle$ in $S_{n-1}^1 - F_e^1 - E_e(u) - \{(u, u_1)\}$. Applying Lemma 3, we can obtain a Hamiltonian path $P(z', y_1)$ in $S_{n-1}^{\{3,\dots,n-1\}} - F_e$. Therefore, $\langle x_1 \rightarrow P(x_1, v') \rightarrow v', v \rightarrow P(v, z) \rightarrow z, z' \rightarrow P(z', y_1) \rightarrow y_1 \rangle$ and $\langle x_2 \rightarrow P(x_2, u) \rightarrow u, u' \rightarrow P(u', y_2) \rightarrow y_2 \rangle$ form two spanning disjoint paths of $S_n - F_e$. as illustrated in Figure 17.

Case 2.2: $x_1, y_1 \in V(S_{n-1}^k)$ for some $2 \leq k \leq n-1$

Without loss of generality, we can assume that $x_1, y_1 \in V(S_{n-1}^2)$ and $f_e^1 \geq f_e^n$.

Case 2.2-1: $f_e^2 + |E_e(y_1)| \leq n-5$

Let $(v, v') \notin F_e$ be an edge for $v \in V(S_{n-1}^2), v' \in V(S_{n-1}^1)$. Applying Lemma 1, we can obtain Hamiltonian paths $P(x_2, v')$ in $S_{n-1}^1 - F_e^1$ and $\langle x_1 \rightarrow P(x_1, y_1) \rightarrow y_1, z \rightarrow P(z, v) \rightarrow v \rangle$ in $S_{n-1}^2 - F_e^2$ for $z' \notin V(S_{n-1}^1)$.

Suppose that $z' \in V(S_{n-1}^n)$. Let (u, w)

4 Conclusion

In this paper, we obtain the following result. There are two spanning disjoint paths for two pair of vertices $x_1, x_2 \in V_1$ and $y_1, y_2 \in V_2$ in S_n for $n \geq 5$ and $F_e \leq n - 4$. The star graph S_n is a 2-pair spanning graph.

References

- [1] SHELDON B. AKERS, "A Group-Theoretic Model for Symmetric Interconnection Networks," *IEEE Transactions on Computers*, pp.555-566, 1989.
- [2] Chun-Nan Hung and Guan-Yu Shi, "Vertex fault tolerance for multiple spanning paths in hypercube," *Processing of the 24rd Workshop on Combinatorial Mathematics and Computational Theory*, pp.241-250, 2007.
- [3] Tseng-Kuei Li, Jimmy J.M. Tan and Lih-Hsing Hsu, "Hyper hamiltonian laceability on edge fault star graph," *Information Sciences*, pp. 59-71, 2004.
- [4] C. K. Lin, H. M. Huanga, and L. H. Hsub, "The super connectivity of the pancake graphs and the super laceability of the star graphs," *Theoretical Computer Science*, pp. 257-271, 2005.
- [5] C. D. Park and K. Y. Chwa, "Hamiltonian properties on the class of hypercube-like networks," *Information Processing Letters*, pp.11-17, 2004.
- [6] Chang-Hsiung Tsai, Jimmy J.M.Tan, Tyne Liang, and Lih-Hsing Hsu, "Fault-tolerant Hamiltonian laceability of hypercubes," *Information Processing Letters*, pp.301-306, (2002).